Lecture 5

Fusion categories: particles and how they fuse

Objectives. *Lectures 1–4 built the language: semisimple categories, tensor products, duals and loops. This lecture puts them together into the setting of anyon models. A fusion category has finitely many particle types. Two particles fuse into a sum of types, counted by multiplicities* N_{ab}^{c}*. States with a fixed total charge form* fusion spaces*. For repeated Fibonacci particles with total charge* τ*, their dimensions are Fibonacci numbers. We also count even numbers of Ising particles with vacuum charge.*

By the end of this lecture you will be able to:

- State the definition of a fusion category and give the fusion rules of Vec_G, $\mathrm{Rep}(S_3)$, Fibonacci, Ising and $\mathrm{SU}(2)_k$.
- Prove that the multiplicities form an associative fusion ring with $N_{ab}^{c} = N_{bc^*}^{a^*}$.
- Compute Frobenius–Perron and quantum dimensions and the global dimension.
- Use the trivalent vertex, the bubble and completeness relations, and count fusion spaces with fusion trees.

5.1 The definition

Definition 5.1 (Fusion category)**.** A *fusion category* is a $\mathbb{C}$-linear semisimple category with finitely many simple objects (Definition 2.11), with a monoidal structure whose tensor product is bilinear on morphisms, which is rigid (Definition 4.1), and whose unit 1 is simple. We take it pivotal (Definition 4.6), as all our examples are; then $X^{**} \cong X$.

The simple objects $a, b, c, \ldots$ are the *particle types*, or anyons; 1 is the vacuum, and the dual a^*, which we also write $\bar{a}$, is the antiparticle. The tensor product of two simple objects decomposes,

$$a \otimes b \cong \bigoplus_{c} N_{ab}^{c}\, c,$$

with *fusion multiplicities* $N_{ab}^{c} = \dim \mathrm{Hom}(a \otimes b, c)$ (Lemma 2.12).

Particle types and state spaces

A particle type is a simple object. A state with prescribed particle types belongs to a vector space of morphisms. For fixed external types $a_1, \dots, a_n$ and total charge c, this is the *fusion space*

$$V_c^{a_1 \cdots a_n} = \mathrm{Hom}(c, a_1 \otimes \cdots \otimes a_n).$$

Choose a bracketing of the tensor product. In a unitary model the dagger gives this space an inner product. The number of independent states is its ordinary integer dimension. The quantum dimension of a simple object, defined by the loop in Definition 4.6, is a different quantity.

The familiar angular-momentum example separates the two kinds of space. Write V_j for the spin-j irreducible representation of SU(2). Adding two and then three spin-$\frac{1}{2}$ particles gives (*QM-I* Thm. 13.2)

$$V_{1/2} \otimes V_{1/2} \cong V_0 \oplus V_1,$$
$$V_{1/2}^{\otimes 3} \cong 2V_{1/2} \oplus V_{3/2}.$$

For the second line, the first pair can have spin 0 or spin 1. Each choice allows total spin $\frac{1}{2}$, so $\mathrm{Hom}_{\mathrm{SU}(2)}(V_{1/2}, V_{1/2}^{\otimes 3})$ has dimension 2. This counts the copies of the spin multiplet. Each copy of $V_{1/2}$ itself has dimension 2. Thus the full total-spin-$\frac{1}{2}$ sector has dimension 4, and the whole three-spin Hilbert space has dimension 8. The fusion space keeps the multiplicity, rather than the magnetic states inside each multiplet.

Rigidity means that every particle can be created from the vacuum together with its antiparticle, by coev_a, and annihilated with it, by ev_a. Rep(SU(2)) has every property but finiteness, since its spins $j = 0, \frac{1}{2}, 1, \dots$ do not stop. The category $\mathrm{Vect}_{\mathrm{fd}}$ is the trivial fusion category, whose only particle is the vacuum.

Example 5.2 (Fusion categories).

- Vec_G^{ω} for a finite group G (Lecture 3): simple objects $g \in G$, $g \otimes h = gh$, and $g^* = g^{-1}$. Every simple object is invertible; such a category is *pointed*.
- $\mathrm{Rep}(G)$ for a finite group (Lectures 2 to 4): the fusion multiplicities are the Clebsch–Gordan multiplicities, computed from characters since $\chi_{V\otimes W} = \chi_V \chi_W$ (*Symmetries* Prop. 8.2), and V^* is the dual representation. For S_3, with simple objects $\mathbf{1}$, sgn, std, all self-dual (*Symmetries* Ex. 8.4):

$$\mathrm{sgn} \otimes \mathrm{sgn} \cong \mathbf{1}, \quad \mathrm{sgn} \otimes \mathrm{std} \cong \mathrm{std},$$
$$\mathrm{std} \otimes \mathrm{std} \cong \mathbf{1} \oplus \mathrm{sgn} \oplus \mathrm{std}.$$

- *Fibonacci*: simple objects $1, \tau$, with

$$\tau \otimes \tau \cong 1 \oplus \tau.$$

- *Ising*: simple objects $1, \sigma, \psi$, with

$$\sigma \otimes \sigma \cong 1 \oplus \psi,$$
$$\sigma \otimes \psi \cong \sigma, \quad \psi \otimes \psi \cong 1.$$

- $\mathrm{SU}(2)_k$: simple objects $j = 0, \frac{1}{2}, \dots, \frac{k}{2}$, with the truncated Clebsch–Gordan rule

$$j_1 \otimes j_2 \cong \bigoplus_{j=|j_1-j_2|}^{\min(j_1+j_2,\, k-j_1-j_2)} j,$$

  in integer steps; without the truncation it is *Symmetries*' Thm. 14.12.

In the last three every simple object is self-dual: $N_{aa}^{1} = 1$ for each a. For Fibonacci, Ising and $\mathrm{SU}(2)_k$ we have so far only the rules; that categories with these rules exist is shown in Lecture 6, by solving the pentagon for Fibonacci and verifying a solution for Ising, and for $\mathrm{SU}(2)_k$ in Lecture 9, from a cited construction.

Two channels for three Fibonacci particles

The Fibonacci rule lists the possible total charges of a pair. The decomposition $\tau \otimes \tau \cong 1 \oplus \tau$ does not by itself choose a state or its amplitudes. If the total charge is fixed to 1, the two-particle fusion space is a line. If the total charge is fixed to τ, it is another line. For three particles with total charge τ, both intermediate pair charges are allowed, so

$$\dim V_{1}^{\tau\tau} = \dim V_{\tau}^{\tau\tau} = 1, \qquad \dim V_{\tau}^{\tau\tau\tau} = 2.$$

A basis for this last space is labelled by whether the first pair has charge 1 or τ. The total charge and all three external particle types stay fixed. A state can have two complex amplitudes on these two intermediate channels. The fusion-tree construction in § 5.5 will realize that basis explicitly.

The quantum dimension computed below is $d_\tau = \varphi$, where $\varphi = (1+\sqrt{5})/2$. This noninteger value does not mean that a single τ carries an ordinary Hilbert space of dimension φ. It describes the growth of the integer dimensions of many-particle fusion spaces. The recurrence proved in § 5.5 and the Binet formula in Sheet 5 make that statement precise.

5.2 Fusion rings

Theorem 5.3 (The fusion ring · *proved given coherence*)**.** *Let $K(C)$ be the free abelian group on the simple objects, with*

$$[a][b] = \sum_{c} N_{ab}^{c}\,[c].$$

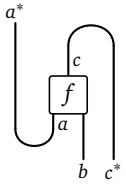
This product is associative with unit $[1]$, and

$$N_{ab}^{1} = \delta_{b,a^*},$$
$$N_{ab}^{c} = N_{bc^*}^{a^*},$$
$$N_{ab}^{c} = N_{b^*a^*}^{c^*}.$$

Proof. Associativity: $(a \otimes b) \otimes c \cong a \otimes (b \otimes c)$, and the multiplicity of e on the two sides is $\sum_x N_{ab}^{x} N_{xc}^{e}$ and $\sum_y N_{bc}^{y} N_{ay}^{e}$. The unit: $1 \otimes a \cong a$. Bending wires (Theorem 4.5) gives $\mathrm{Hom}(a \otimes b, 1) \cong \mathrm{Hom}(a, b^*)$, which by Schur is $\mathbb{C}$ if $a \cong b^*$ and 0 otherwise (b^* is simple, since the rotated box is a bijection $\mathrm{End}(b) \to \mathrm{End}(b^*)$); since $b^{**} \cong b$, $a \cong b^*$ exactly when $b \cong a^*$. For the second identity, bend c down and then a across, using its right dual (also a^*, since C is pivotal):

$$\begin{aligned}\mathrm{Hom}(a \otimes b, c) &\cong \mathrm{Hom}(a \otimes b \otimes c^*, 1)\\ &\cong \mathrm{Hom}(b \otimes c^*, a^*),\end{aligned}$$

and compare dimensions. In pictures, f goes to



whose inverse bends the two strands back. For the third, the rotated box gives $\mathrm{Hom}(a\otimes b, c) \cong \mathrm{Hom}(c^*, (a\otimes b)^*)$, and $(a\otimes b)^* \cong b^*\otimes a^*$ (Proposition 4.3); by semisimplicity the multiplicity of c^* in $b^*\otimes a^*$ is the dimension of either Hom space between them. ∎

For Fibonacci the two bracketings of three τ's give

$$(\tau\otimes\tau)\otimes\tau \cong (1\oplus\tau)\otimes\tau \cong 1\oplus 2\tau,$$
$$\tau\otimes(\tau\otimes\tau) \cong \tau\otimes(1\oplus\tau) \cong 1\oplus 2\tau.$$

The ring records that the two sides are isomorphic, but not which isomorphism the associator is; that is the content of Lecture 6.

The *fusion matrix* N_a is the matrix of left multiplication by $[a]$, $(N_a)_{cb} = N_{ab}^c$. For Fibonacci and Ising, in the bases $1, \tau$ and $1, \sigma, \psi$,

$$N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad N_\sigma = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

and associativity reads $N_aN_b = \sum_c N_{ab}^c N_c$. The matrices need not commute: in Vec_G for $G = S_3$, $N_gN_h = N_{gh}$. The rings of our anyon models are commutative, because $a\otimes b \cong b\otimes a$ once there is a braiding (Lecture 7). A ring with a basis of this kind, non-negative integer structure constants and the properties of the theorem is a *fusion ring*.

Different categories can share a ring. For $k = 2$ the truncated rule gives $\frac{1}{2}\otimes\frac{1}{2} \cong 0\oplus 1$, $\frac{1}{2}\otimes 1 \cong \frac{1}{2}$ and $1\otimes 1 \cong 0$: under $0, \frac{1}{2}, 1 \mapsto 1, \sigma, \psi$ this is the Ising ring, though Lecture 6 shows that the two categories differ. For $k = 3$, $1\otimes 1 \cong 0\oplus 1$, since the truncation removes $j = 2$; the objects 0 and 1 fuse like 1 and τ.

The fusion ring does not determine the category. Vec_G and Vec_G^ω for $G = \mathbb{Z}_2$ have the same ring and are not equivalent (Lecture 3). For groups: the dihedral group of order 8 and the quaternion group have the same character table (*Symmetries* Ex. 5.15, which calls the dihedral group D_4), hence the same fusion ring, yet their representation categories are not equivalent as fusion categories (Calaque–Etingof, Cor. 3.12 · *cited*).

5.3 FPdim; the canonical spherical structure

The fusion matrices determine the Frobenius–Perron dimensions.

Theorem 5.4 (Frobenius–Perron dimension · *sketched given coherence*)**.** *For each simple a, the matrix N_a has non-negative entries and a largest real eigenvalue $\mathrm{FPdim}(a) \geq 1$ with a positive eigenvector. The map $a \mapsto \mathrm{FPdim}(a)$ is multiplicative,*

$$\mathrm{FPdim}(a)\,\mathrm{FPdim}(b) = \sum_c N_{ab}^c\,\mathrm{FPdim}(c),$$

and it is the only such function that is positive on every simple object (EGNO Prop. 3.3.6).

Sketch. Let $R = \sum_a [a]$, and let M be the matrix of right multiplication by R. Its entries $\sum_a N_{ba}^c = \sum_a \dim \mathrm{Hom}(a, b^* \otimes c)$ are positive, since $b^* \otimes c \neq 0$ (1 is a summand of $b \otimes b^*$, so c is one of $b \otimes b^* \otimes c$). By the Perron–Frobenius theorem M has a positive eigenvector v, unique up to scale, for its largest eigenvalue. Left and right multiplications commute, by associativity, so each N_a maps this eigenline to itself: $N_a v = \lambda_a v$, and $\lambda_a > 0$ because N_a is non-negative and non-zero. In fact $\lambda_a \geq 1$: every column of N_a has an entry at least 1, since $a \otimes b \neq 0$, so the entries of $N_a v$ add up to at least those of v. A non-negative matrix with a positive eigenvector has that eigenvalue as its largest, so $\lambda_a = \mathrm{FPdim}(a)$; applying $N_a N_b = \sum_c N_{ab}^c N_c$ to v gives multiplicativity. Conversely, a positive multiplicative f makes $(f(c^*))_c$ a positive eigenvector of M, so it is proportional to v; with $f(1) = 1$ this pins f down. ∎

For Fibonacci N_τ has eigenvalues $\varphi = (1+\sqrt{5})/2$ and $-1/\varphi$, so $\mathrm{FPdim}(\tau) = \varphi$; for Ising $\mathrm{FPdim}(\sigma) = \sqrt{2}$ and $\mathrm{FPdim}(\psi) = 1$; in $\mathrm{Rep}(G)$ it is the dimension. Uniqueness can be seen directly for Fibonacci: a multiplicative function has $d_\tau^2 = 1 + d_\tau$, whose only positive root is φ. The other root $-1/\varphi$ returns in Lecture 6 as the Yang–Lee category. For $\mathrm{SU}(2)_k$ (Sheet 5),

$$\mathrm{FPdim}(j) = \frac{\sin\big((2j+1)\pi/(k+2)\big)}{\sin\big(\pi/(k+2)\big)};$$

for $k = 2$ it gives $1, \sqrt{2}, 1$, Ising's values, and for $k = 3$ it gives $\mathrm{FPdim}(\frac{1}{2}) = \mathrm{FPdim}(1) = \varphi$.

The quantum dimension d_a of Lecture 4 is defined by a loop and depends on the pivotal structure. It too is multiplicative: $d_a d_b = d_{a\otimes b}$ (Theorem 4.7), and the trace is additive on the direct sum $a \otimes b \cong \bigoplus_c N_{ab}^c\, c$, so $d_a d_b = \sum_c N_{ab}^c d_c$. A fusion category is *pseudo-unitary* if $\sum_a d_a^2 = \sum_a \mathrm{FPdim}(a)^2$ for some spherical structure (EGNO Def. 9.4.4).

A fusion category is *unitary* if it has an antilinear dagger (§ 4.5) with $(f \otimes g)^\dagger = f^\dagger \otimes g^\dagger$ and unitary associators and unitors. For every simple a and object X, require the formula

$$\langle v, w \rangle\ \mathrm{id}_a = v^\dagger w, \qquad v, w \in \mathrm{Hom}(a, X),$$

to define a positive inner product. Unitary fusion categories are pseudo-unitary (EGNO Rem. 9.4.7 · *cited*).

> **Theorem 5.5** (The canonical spherical structure · *cited*). *A pseudo-unitary fusion category has a unique spherical structure with $d_a = \mathrm{FPdim}(a)$ for every simple a (EGNO Prop. 9.5.1).*

All the anyon numerics of the course use this canonical structure (so $d_\tau = \varphi$, $d_\sigma = \sqrt{2}$). Other pivotal structures exist: on Vec_G for $G = \mathbb{Z}_2$ the choice $a_s = -\mathrm{id}$ of Lecture 4 gives $d_s = -1$. The *global dimension* is

$$\dim C = \sum_a d_a^2,$$

with the canonical d: $|G|$ for Vec_G and for $\mathrm{Rep}(G)$ (*Symmetries* Thm. 5.11), $2+\varphi$ for Fibonacci, 4 for Ising.

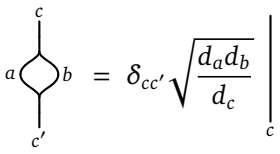
5.4 Vertices, bubbles, completeness

From here on the categories are unitary, with the canonical spherical structure (so every $d_a > 0$), and *multiplicity-free*, every $N_{ab}^c \in \{0, 1\}$, as all the course's anyon models are. Then

$\mathrm{Hom}(c, a \otimes b)$ is a line when $N_{ab}^{c} = 1$, and we choose a basis vector, the *splitting vertex* ψ_c^{ab}, drawn with c below; its dagger, the *fusion vertex* $\psi_c^{ab\dagger}$, is the picture reflected top to bottom:



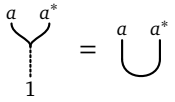
The vertex is fixed up to a scalar, and we fix its size by the *isotopy-invariant normalisation* (Bonderson (2.14)–(2.19); Kitaev eqs. 195–196): a fusion vertex on a splitting vertex is



(5.1)

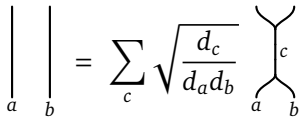
This can be arranged: by Schur the left side is a multiple of id_c, zero unless $c = c'$, and positive since it has the form $\psi^\dagger\psi$; rescaling ψ by a positive number sets it. A phase remains free, the gauge of Lecture 6.

The vacuum channels are fixed by Lecture 4. Choosing $\psi_a^{1a} = \psi_a^{a1} = \mathrm{id}_a$ satisfies (5.1). For $a \otimes a^* \to 1$, the bubble asks $\psi^\dagger\psi = \sqrt{d_a d_{a^*}} = d_a$. Choose the duality maps *standard*: each cap is the dagger of a cup, $\mathrm{coev}_a^\dagger = \widetilde{\mathrm{ev}}_a$, as in $\mathrm{Hilb}_{\mathrm{fd}}$ (Theorem 4.10); this can be done with the canonical dimensions (Kitaev eqs. (190)–(191) · *cited*). Then $\mathrm{coev}_a^\dagger\,\mathrm{coev}_a$ is the loop d_a, and we may take the cup itself:



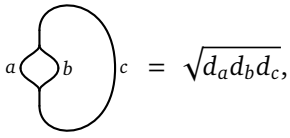
(5.2)

For a self-dual a this identifies a^* with a; when $\kappa_a = -1$, as for spin $\frac{1}{2}$ in $\mathrm{SU}(2)_k$, the cup carries the flag of Lecture 4. Since $a \otimes b$ is the direct sum of its channels, the identity of $a \otimes b$ is the sum of the projections onto them:



(5.3)

the coefficient making each term an idempotent by (5.1). Closing the c strand of (5.1) into a loop gives the *theta graph*,



(5.4)

symmetric in a, b, c. Taking the trace of (5.3), the left side becomes two loops, $d_a d_b$; by cyclicity of the trace each term on the right becomes $\sqrt{d_c/d_a d_b}$ times a theta graph, that is d_c. So

$$d_a d_b = \sum_c N_{ab}^{c}\, d_c,$$

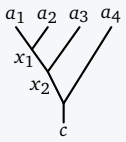
recovering the multiplicativity of § 5.3.

In $\mathrm{Rep}(S_3)$, for std ⊗ std, a splitting vertex is an isometric intertwiner $c \to \mathrm{std} \otimes \mathrm{std}$ times $(4/d_c)^{1/4}$, and both relations can be checked as matrix identities. With std real orthogonal on an orthonormal basis e_1, e_2, the vacuum vertex is $1 \mapsto e_1 \otimes e_1 + e_2 \otimes e_2$, which is $\mathrm{coev}_{\mathrm{std}}$ under the intertwiner $\mathrm{std}^* \to \mathrm{std}$, $e^i \mapsto e_i$, and its bubble is $2 = d_{\mathrm{std}}$. Kitaev's App. E.1 uses unit-norm vertices instead, $\psi^\dagger\psi = \mathrm{id}$, with no square roots in completeness; the two conventions differ by $(d_c/d_a d_b)^{1/4}$ per vertex, and a formula must not mix them.

5.5 Fusion spaces

The fusion space $V_c^{a_1\cdots a_n}$ was defined at the start of the lecture. We now construct its basis by decomposing the tensor product one pair at a time. As in the spin example, the intermediate labels count multiplicities at fixed total charge.

Theorem 5.6 (Fusion trees). *Fix the left-associated bracketing. The* fusion trees *of this shape with root c and leaves $a_1, \dots, a_n$, labelled by intermediate simple objects $x_1, \dots, x_{n-2}$ with every vertex allowed, form a basis of $V_c^{a_1\cdots a_n}$:*



So

$$\dim V_c^{a_1\cdots a_n} = \sum_x N_{a_1a_2}^{x_1} N_{x_1a_3}^{x_2} \cdots N_{x_{n-2}a_n}^{c}.$$

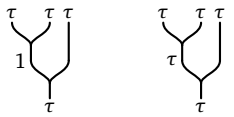
Proof. For $n = 2$ the space is a line or zero. For $n > 2$, decompose $a_1 \otimes \cdots \otimes a_{n-1}$ into simple objects by induction: by semisimplicity,

$$\begin{aligned}\mathrm{Hom}(c, a_1 \otimes \cdots \otimes a_n) &\cong \bigoplus_x \mathrm{Hom}(c, x \otimes a_n) \\ &\quad \otimes \mathrm{Hom}(x, a_1 \otimes \cdots \otimes a_{n-1}),\end{aligned}$$

and a basis is the vertex $\psi_c^{xa_n}$ followed by the trees of the second factor. ∎

Bracketing the leaves differently gives another basis of the same space; the change of basis is the F-move of Lecture 6.

Three τ's with total charge τ have two trees, with $x_1 = 1$ and $x_1 = \tau$:



In general each intermediate label of n τ's is 1 or τ, and a 1 cannot follow a 1. Write f_n for $\dim V_\tau^{\tau\cdots\tau}$ with n leaves; the last vertex is $\tau \otimes \tau \to \tau$ or $1 \otimes \tau \to \tau$, and a 1 after $n-1$ leaves must come from τ after $n-2$, so

$$f_n = f_{n-1} + f_{n-2}, \qquad f_1 = f_2 = 1.$$

The dimensions are $1, 1, 2, 3, 5, 8, \dots$, the Fibonacci numbers, and they grow like φ^n (Sheet 5): a large degenerate space, protected by topology. For Ising, $2m$ copies of σ with total charge 1 have 2^{m-1} states. For $m = 2$ the trees have $x_1 \in \{1, \psi\}$ and then $x_2 = \sigma$: four σ's carry one qubit.

Summary.

- *Handed on:* a fusion category is semisimple, rigid and pivotal, with finitely many simples, a simple unit and bilinear $\otimes$ (Definition 5.1); Fibonacci, Ising, $\mathrm{SU}(2)_k$, Vec_G^ω and $\mathrm{Rep}(G)$ are the examples.
- *The fusion ring:* N_{ab}^c form an associative ring with $N_{ab}^1 = \delta_{b,a^*}$ and $N_{ab}^c = N_{bc^*}^{a^*}$ (Theorem 5.3); it does not determine the category.
- *Dimensions:* FPdim is the Perron–Frobenius eigenvalue, φ for τ and $\sqrt{2}$ for σ; for a pseudo-unitary C with its canonical spherical structure $d = \mathrm{FPdim}$, and $\dim C = \sum_a d_a^2$ (Theorems 5.4 and 5.5).
- *Pictures:* the trivalent vertex with the bubble $\sqrt{d_a d_b / d_c}$ and completeness (5.1), (5.3); admissibly labelled trees of one fixed shape form a basis, with Fibonacci numbers for τ (Theorem 5.6).
- *Next:* Lecture 6 computes the associator on fusion trees, the F-symbols, and solves the pentagon for Fibonacci.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: fusion categories, Def. 4.1.1, p. 65; based and fusion rings, Defs. 3.1.3, 3.1.7 and Prop. 3.1.6, pp. 49–50; the Frobenius–Perron dimension, Def. 3.3.3 and Prop. 3.3.6, p. 53; the Yang–Lee (Fibonacci), Ising and Verlinde rings, Exs. 3.1.9 and 4.10.6, pp. 51 and 79; pseudo-unitary categories, Def. 9.4.4, p. 283, and unitary ones, Rem. 9.4.7, pp. 283–284; the canonical spherical structure, Prop. 9.5.1, p. 284. Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438, App. E: fusion rules, p. 67, unit-norm vertices, eqs. 174–176, p. 68, the standard cups and caps, eqs. (190)–(191), p. 76, the isotopy-invariant normalisation, eqs. 195–196, p. 77, quantum dimensions as Perron–Frobenius eigenvalues, pp. 79–80. Bonderson, PhD thesis (Caltech, 2007), (2.1)–(2.19), pp. 13–19. Simon, *Topological Quantum*, Ch. 8, pp. 93–102 (Fibonacci counting, p. 98). Calaque and Etingof, *Lectures on tensor categories*, Cor. 3.12, p. 24.